READING GROUP ON GEOMETRIC MOTIVIC INTEGRATION

KU LEUVEN, SPRING 2026

This reading group covers *geometric motivic integration* and related topics, including arc and jet spaces, motivic measure, motivic zeta functions, and the monodromy conjecture. We start with some motivation and history (such as the connection to p-adic integration). The goal of this reading group is to understand motivic integration (starting from the basics), and to define the motivic zeta function and state the motivic monodromy conjecture. In the last talks, we discuss the monodromy conjecture in some specific settings (such as the plane curves) and connections to other problems. Below is a list of proposed topics for the talks. Each talk is expected to last approximately one hour. Talks (5), (8), and (9) are optional, depending on time and interest.

The reading groups will take place every Thursday at 14:30 in room 02.18. The first session will be held on Thursday, February 19th.

Organisers : Lise Fonteyne, Loïs Faisant

(0) **Motivation and introduction – Lise – 19/02/2026 – [Notes]**

(1) **Pre-history: the p-adic setting – Loïs – 26/02/2026 – [Notes]** – [VS21, §1]
We start with an introduction to p-adic integration, which serves as an inspiration for motivic integration. Indeed, there are many analogies between the two integration theories.
- Definition of the p-adic Igusa zeta function [VS21, §1.5]
- Rationality: Igusa's theorem [VS21, Theorem 1.46] and [Nic10, Theorem 2.7]
- Sketch of the proof (treating technical ingredients as black boxes)
- Denef's formula: statement only, see [VS21, Remark 1.48] and [Nic10, §2.5]
- Emphasis on general ideas, strategies, and context

(2) **Geometric motivic integration, Part I – Sebastian – 12/03/2026 – [Notes]** [VS21, §2.1–2.3] (or [Nic10, §4.3–4.4], [Vey06, §2–3])
In the first talk about motivic integration, we cover the basics needed to define motivic integrals. We discuss the domain of integration, i.e. *arc spaces* $\mathbb{C}[[t]]$ (t-adic spaces), and the motivic measure, based on a *universal additive invariant of varieties*, i.e. the Grothendieck ring of complex varieties $K_0(\mathrm{Var}_{\mathbb{C}})$.
- Arc and jet spaces: definitions and basic properties (e.g. [VS21, Proposition 2.17])
- Motivic measure: first on stable subsets, then on cylinders [VS21, Theorem 2.33]
- Closed subsets have measure zero [VS21, Proposition 2.34]
- Discussion of [VS21, Remark 2.35]

(3) **Geometric motivic integration, Part II – Marko – 19/03/2026** [VS21, §2.4–2.6] (or [Nic10, §4.4–4.5], [Vey06, §3–5])
In this talk, we discuss the main topic of this reading course: motivic integration. This integration theory comes with a *change of variables formula*, where the "jacobian" is expressed in terms of the contact order of arcs along divisors. Kontsevich developed this theory to study the relation between additive invariants of varieties. We briefly take a look at his theorem. Finally, we discuss selected results for computations using an (embedded) resolution of singularities.
- Definition of motivic integration [VS21, Definition 2.36]
- Contact order of an arc along a divisor and the change of variables formula [VS21, §2.4.1–2.4.2]
- Application to Kontsevich's theorem [VS21, §4.5]
- Discussion of selected results from [VS21, §4.6], e.g. Theorem 2.55 and Corollary 2.57

(4) **Motivic zeta function – Shi Quan – 26/03/2026 – [Notes]** [VS21, §3.2] (or [Nic10, §5], [Vey06, §6])
Now, we are ready to define the motivic zeta function, which is a motivic integral. We revise embedded resolutions and their numerical data associated, which pops up in the Denef-Loeser formula. Lastly, we take a look at the p-adic Igusa zeta function and the topological zeta function, and how they are specializations of the motivic zeta function.

- Definition of the motivic zeta function [VS21, Definition 3.17]
- Review of embedded resolutions of singularities and numerical data
- Statement of Theorem 3.18 and Corollary 3.19 from [VS21] (uses an embedded resolution)
- Specializations of the motivic zeta function to the p-adic Igusa zeta function and the topological zeta function

(5) **The monodromy conjecture – Lise – 02/04/2026** [Vey25]
In this talk, we start by stating the monodromy conjecture in the p-addic setting, followed by the motivic and topological versions. We take a look at the proof for plane curves. Next, we go one step further, by looking at the generalized conjecture for curves on singular surfaces.
- Monodromy conjecture in the p-adic setting [Vey25, §2.4]
- Motivic and topological versions of the conjecture
- Proof for plane curves [Vey25, §4]
- What happens for curves on singular surfaces?

(6) **A birational zeta function – Tom – 23/04/2026**

(7) **Connections with cohomology theories – Loïs – 30/04/2026**
The goal of this talk is to present and draw a parallel between different versions of the Milnor fibre and nearby cycles appearing in topology, motivic integration and motives theory.
- Motivic nearby cycles and monodromy [Azo25, §1 and §4]
- For connection, see [Nic10, §5.4]

(8) **The monodromy conjecture and higher dimension – Marko – 07/05/2026** [Vey25, §5]
If there is interest, we can discuss the monodromy conjecture in higher dimensions.
- Are there any strategies? [Vey25, §5.1]
- What is known in higher dimensions? [Vey25, §5.2]

(9) **Generalizations of the monodromy conjecture – Lise – 21/05/2026** [Vey25, §6]
If there is interest, we can discuss some generalizations of the monodromy conjecture, such as:
- Mappings/ideals, zeta functions with differential forms, multivariate zeta functions, etc.

Version of March 25, 2026.

References

[Azo25] Azouri, R. (2025). *Motivic nearby cycles and their monodromy at a singular point.* https://arxiv.org/pdf/2511.07668 2

[Nic10] Nicaise, J. (2010). An introduction to p-adic and motivic zeta functions and the monodromy conjecture. *Algebraic and Analytic Aspects of Zeta Functions and L-functions*, 21, 115–140. 1, 2

[Vey06] Veys, W. (2006). Arc spaces, motivic integration and stringy invariants. *Advanced Studies in Pure Mathematics* 43, 529–572. 1

[Vey25] Veys, W. (2025). Introduction to the monodromy conjecture. In *Handbook of Geometry and Topology of Singularities VII*, 721–765. Springer. 2

[VS21] Viu-Sos, J. (2021). An introduction to p-adic and motivic integration, zeta functions and invariants of singularities. *Contemporary Mathematics*, 778. 1, 2